Activity 04: (Laboratory Exercises,Lists Set 01)

1. print all elements of a list ?-print_list([a,b,c]). a b c print_list([]):-nl. %nl = newline print_list([H|T]):-write(H),write(' '),print_list(T).

2. reverse all elements of a list ?-reversex([a,b,c],X). X=[c,b,a] addtoend(H,[],[H]). addtoend(X,[H|T],[H|T1]):-addtoend(X,T,T1). reversex([],[]). reversex([H|T],Y):-reversex(T,T1), addtoend(H,T1,Y).

3. create list ?-create_list(5,12,S). S=[5,6,7,8,9,10,11,12] create_list(X,X,[X]). create_list(A,X,[A|T]):- AA is A+1, create_list(AA,X,T).

4. mean value [1,2,3,4,5] => 3 sum_list([],0,0). sum_list([H|T],Length,Sum):-sum_list(T,L1,S1), Length is L1+1, Sum is S1+H. mean(L,M):-sum_list(L,Length,Sum), M is Sum/Length.

5. detect whether list contains a number [a,b,c,d,e,1,f] => T numberinlist([]):-fail. numberinlist([X|T]):-number(X). numberinlist([X|T]):-numberinlist(T).

6. increment elements of list [5,6,7,8] => [6,7,8,9] increment([],[]). increment([H|T],[X|Y]):-increment(T,Y),X is H+1.

7. factorial function factorial(0,1). factorial(N,X):-N>0, N1 is N-1, factorial(N1,S), X is S*N.

8. implement append function [a,1,2,b,c], [b,c,d,e] => [a,1,2,b,c,b,c,d,e] appendx([],A,A). appendx([H|T],A,[H|U]):-appendx(T,A,U).

9. encapsulate list elements [a,b,1,d,e] => [[a],[b],[1],[d],[e]] encapsulate([],[]). encapsulate([H|T],[[H]|Y]):-encapsulate(T,Y).

10. insert zeros [1,2,3,4,5] => [1,0,2,0,3,0,4,0,5,0] insert_zeros([],[]). insert_zeros([H|T],[H,0|Y]):-insert_zeros(T,Y).

11. clone list [g,6,7] => [[g,6,7][g,6,7]] clone_list(T,[T,T]).

12. modify list element ?-modify_list([m,o,d,i,f,y,e,t], 6, i,Y). Y=[m,o,d,i,f,y,i,t], ?-modify_list([m,o,d],6,i,Y). Y=[m,o,d] modify_list([],N,X,[]). modify_list([H|T],0,X,[X|T]). modify_list([H|T],N,X,[H|Y]):-N>0, N1 is N-1, modify_list(T,N1,X,Y).

13. rotate list [1,2,3,4,5] => [2,3,4,5,1] addtoend(H,[],[H]). addtoend(X,[H|T],[H|T1]):-addtoend(X,T,T1). rotate_list([H|T],L1):-addtoend(H,T,L1).

14. Fibonacci numbers: print nth Fibonacci number. fib(1,1). fib(2,1). fib(N,F):- N>2, N1 is N-1, fib(N1,F1), N2 is N-2, fib(N2,F2), F is F1+F2.

15. generate random (values 0-9) square matrix: [[2,4,5],[1,0,3],[9,3,2]]. Inner elements

represent matrix rows.

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random10(N):-frandom(X), Y is X*10, fix(Y,N).
rand_row(0,[]).
rand_row(N,[H|T]):-N>0,random10(H),N1 is N-1,
rand_row(N1,T).
square_matrix_rand(N,S):-smr(N,N,S).
smr(N,0,[]).
smr(N,X,[R|T]):-N>0, rand_row(N,R),X1 is X-1,smr(N,X1,T).
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